Flexible Attention-Based Multi-Policy Fusion for Efficient Deep Reinforcement Learning

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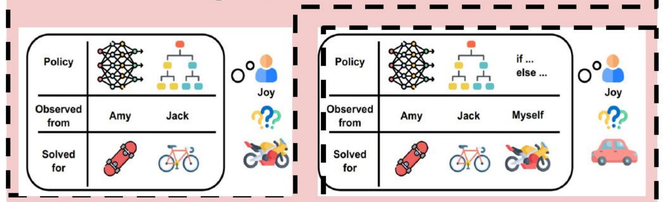
Why Knowledge-Grounded Reinforcement Learning?

- What learning capabilities do humans possess, yet RL agents still missing? [1,2]
- **Knowledge-Acquirable**: Develop unique policies by observing.
- **Compositional**: Combine policies from multiple sources to form a knowledge set.

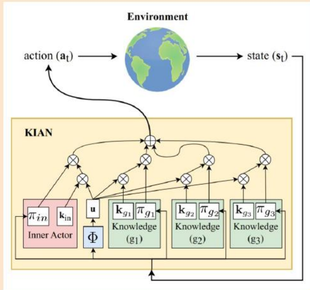


- **Incremental**: Arbitrarily remove outdated policies or add new ones, no need to relearn how to navigate the entire knowledge set from scratch.
- **Sample-Efficient**: Require fewer interactions.
- **Generalizable**: Be able to apply previously learned policies to unseen tasks.
- **KGRL**: An RL paradigm aiming to learn with external policies and achieve the five properties above.

Method: Knowledge-Inclusive Attention Network (KIAN)

- **KIAN:** An actor that separates knowledge representations from their selection mechanism.

Model Architecture:
- Inner Actor
- Knowledge Keys
- Query



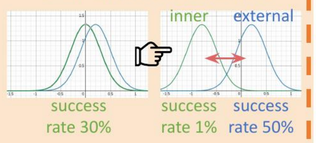
- Naive policy fusion leads to challenges:

$$\pi(\cdot|\mathbf{s}_t) = \hat{w}_{in}\pi_{in}(\cdot|\mathbf{s}_t) + \Sigma_{j=1}^{n}\hat{w}_{g_j}\pi_{g_j}(\cdot|\mathbf{s}_t)$$

- **Discrete vs Continuous** action space
- **Entropy Imbalance** in maximum entropy KGRL

P1: always inner actor

P2: distant means



Discrete Action Policy

$$\pi(\cdot|\mathbf{s}_t) = \hat{w}_{t,in}\pi_{in}(\cdot|\mathbf{s}_t) + \Sigma_{j=1}^{n}\hat{w}_{t,g_j}\texttt{softmax}(\beta_{t,g_j}\pi_{g_j}(\cdot|\mathbf{s}_t))$$

$$w_{t,g_j} = \Phi(\mathbf{s}_t)\cdot\mathbf{k}_{g_j}/\beta_{t,g_j} \quad \beta_{t,g_j} = \|\Phi(\mathbf{s}_t)\|_2\|\mathbf{k}_{g_j}\|_2$$

Continuous Action Prediction

$$\pi_{in}(\cdot|\mathbf{s}_t) = \mathcal{N}(\boldsymbol{\mu}_{t,in}, \boldsymbol{\sigma}^2_{t,in}), \pi_{g_j}(\cdot|\mathbf{s}_t) = \mathcal{N}(\boldsymbol{\mu}_{t,g_j}, \boldsymbol{\sigma}^2_{t,g_j})$$

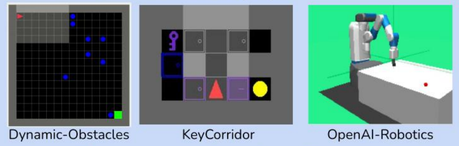
$$e \sim \texttt{gumbel_softmax}([\hat{w}_{t,in}, \hat{w}_{t,g_1}, \ldots, \hat{w}_{t,g_n}]^\top) \quad \mathbf{a}_t \sim \pi_e(\cdot|\mathbf{s}_t)$$

Continuous Probability Distribution

$$\pi(\mathbf{a}_t|\mathbf{s}_t) = \hat{w}_{in}\pi_{in}(\mathbf{a}_{t,in}|\mathbf{s}_t) + \Sigma_{j=1}^{n}\hat{w}_{g_j}\pi_{g_j}(\boldsymbol{\mu}_{t,g_j}|\mathbf{s}_t)$$

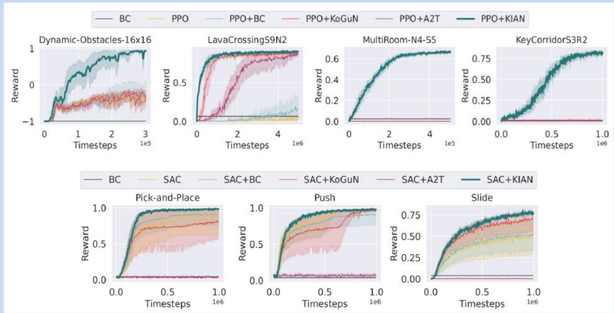
How well are the KGRL properties satisfied?

- Environments:
  - Discrete decision making (MiniGrid)
  - Continuous control (OpenAI-Robotics)



- RL algorithms: proximal policy optimization (PPO) and soft actor critic (SAC)
- **Main Results:**

Sample Efficiency & Compositional



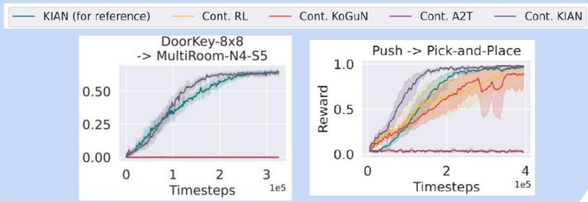
Generalizability

- Zero-shot transfer policies from simple to more complex tasks

Train in / Test in	Empty-Random-5x5 6x6	8x8	16x16	DoorKey-5x5 8x8	16x16	Push 5x	10x	Slide 5x	10x	Pick-and-Place 5x	10x
RL	0.88	0.71	0.45	0.29	0.08	0.87	0.52	0.45	0.17	0.34	0.27
RL+BC	0.87	0.60	0.24	0.40	0.09	0.89	0.60	0.44	0.16	0.34	0.30
KoGuN [3]	0.94	0.83	0.53	**0.77**	0.35	0.63	0.43	**0.55**	**0.18**	0.32	0.24
A2T [4]	0.92	0.78	0.51	0.53	0.11	0.03	0.05	0.00	0.01	0.01	0.06
KIAN (ours)	**0.96**	**0.91**	**0.93**	0.76	**0.42**	**0.93**	**0.70**	0.42	0.15	**0.92**	**0.72**

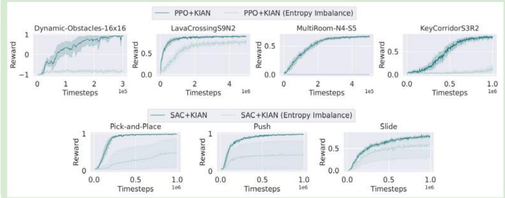
Compositional & Incremental

- Combine previously learned knowledge keys and inner policies to learn a new task



How entropy imbalance affects?

- Addressing entropy imbalance is important!



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